Cloud Computing Tutorials -4 & 5

CloudSim Stimulation

Observation:

Date:

Tutorial - 4 and 5
CloudSim Simulation

Q. what is cloudSim?
CloudSim is a simulation tool that helps researchers and students study how cloud computing environment work without needing real cloud infrastructure. It is used to model and test cloud computing systems, including virtual machines (vms), data centre, and scheduling policies, before deploying them in real-world scenerios.

Q what does cloudSim do?
CloudSim allows users to:
1. Simulate cloud infrastructure
2. Testing scheduling & Resource allocation
3. Evaluate performance
4. optimize cloud services.

CloudSim is a framework for simulating cloud computing environment and evaluating different cloud strategies. It is implemented in Java and requires Java and eclipse for execution.

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Prerequisites : JDK 8 or higher
Eclipse IDE
CloudSim source code
Apache common math Library

Add CloudSim source code into eclipse
- manually copy and paste code into eclipse project
1. open eclipse and create a new java project
2. paste CloudSim source code
3. refresh the project

Add required libraries
- add external JAR files

Run a CloudSim Example
Eg:- CloudSimExample.java in source
Flow output will display in console

output :-
starting CloudSimExample...
Initializing
starting cloudsim version 3.0
Datacenter_0 is starting
⋮
Simulation completed
output status → Success
CloudSim Example! Finished!

CloudSim Simulation:

```
C:\Users\Admin>java --version
java 23.0.1 2024-10-15
Java(TM) SE Runtime Environment (build 23.0.1+11-39)
Java HotSpot(TM) 64-Bit Server VM (build 23.0.1+11-39, mixed mode, sharing)

C:\Users\Admin>
```

Fig 1.1 : JDK Installation Verfication

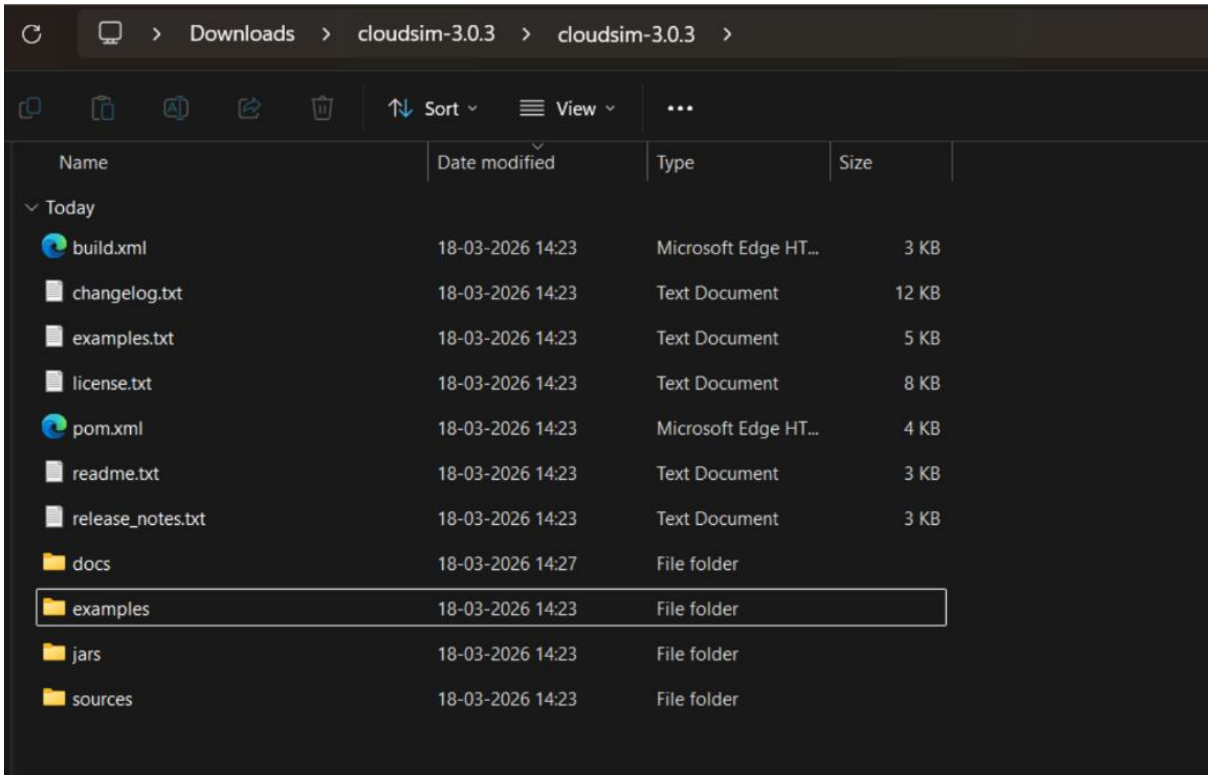

Fig 1.2 : cloudsim zip folder extraction

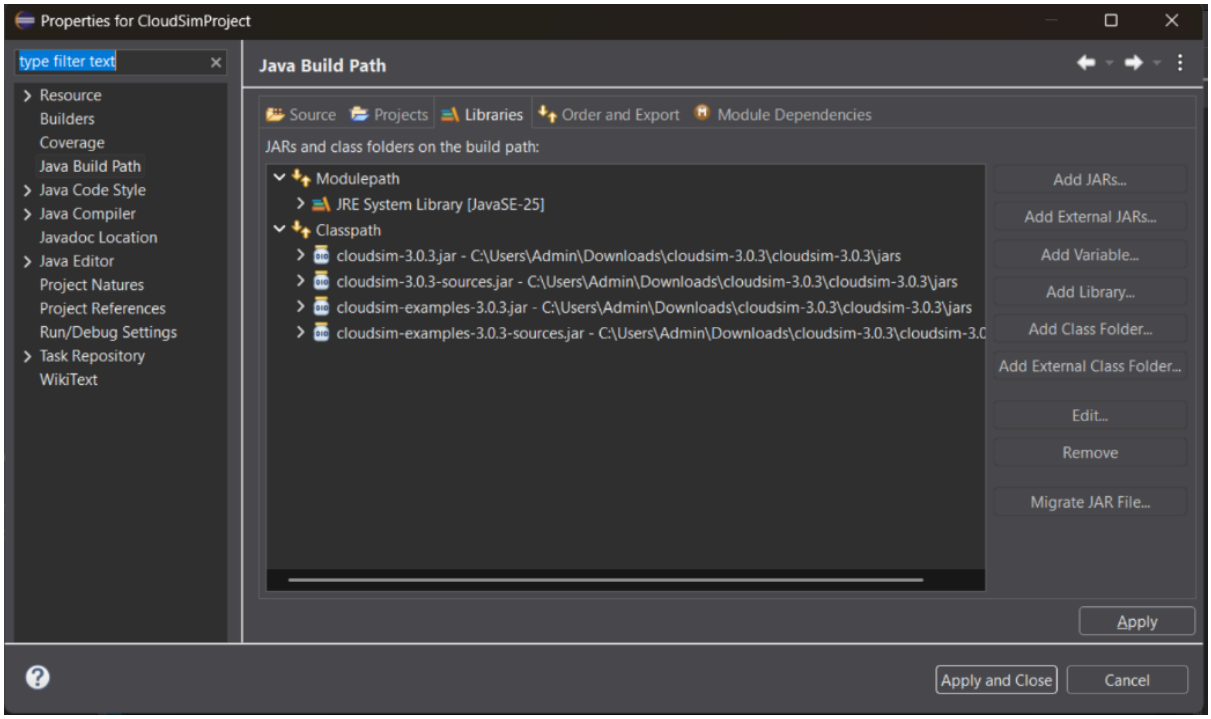

Fig 1.3 : CloudSim Project properties

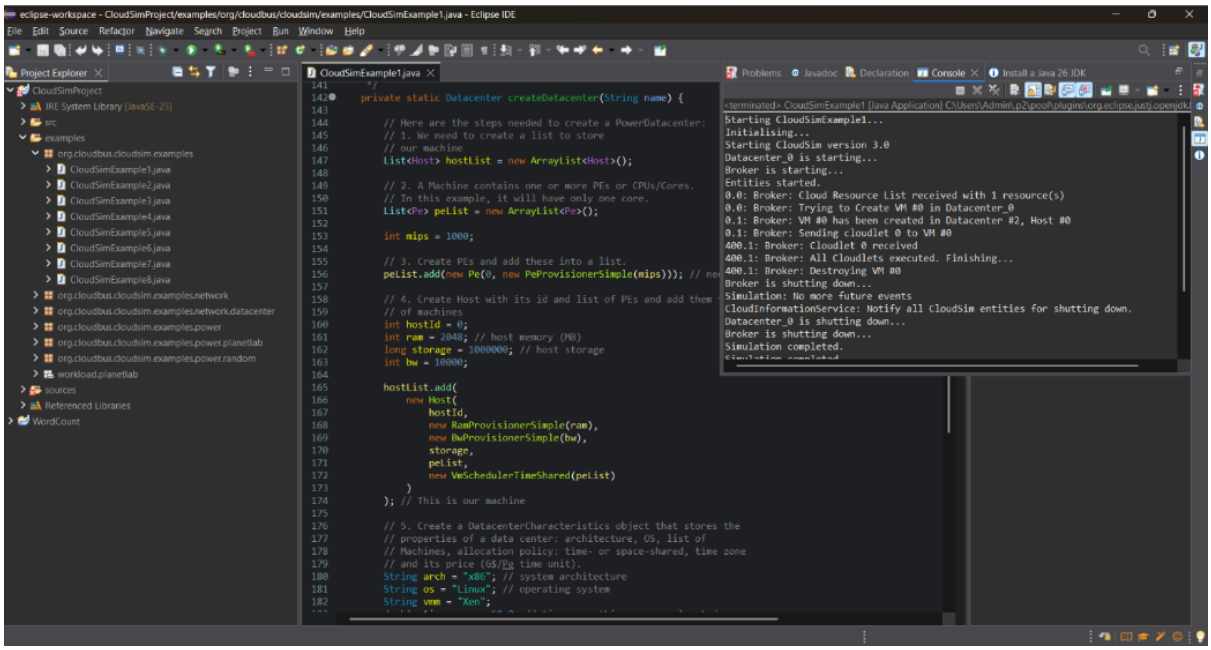

Fig 1.4: Running CloudSim project – CloudSim example -1 .java program

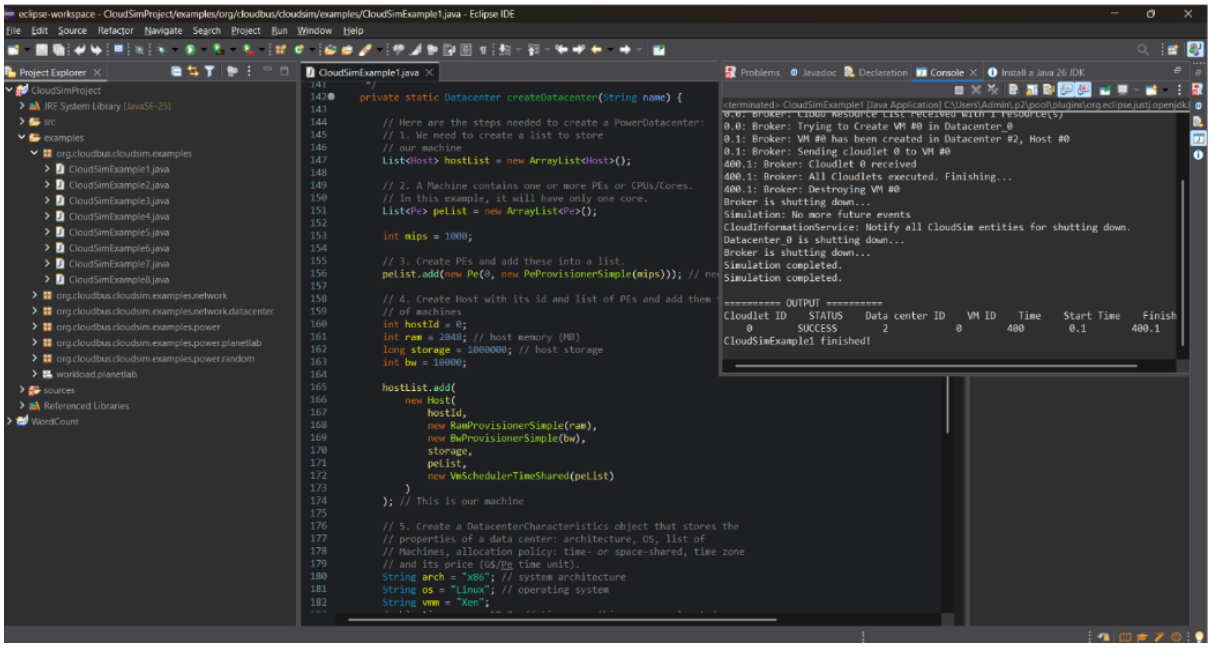

Fig 1.5 : Stimulation successful